

2.2 Addition Principle If an operation can be performed in ' m ' different ways and another operation, which is independent of the first operation, can be performed in ' n ' different ways. Then either of the two operations can be performed in $(m + n)$ ways. This can be extended to any finite number of independent operations.

Example 4: A college offers 6 courses in the morning and 4 in the evening. Find the number of ways a student can select exactly one course, either in the morning or in the evening.

Solution: The college has 6 courses in the morning out of which the student can select one course in 6 ways.

In the evening the college has 4 courses out of which the student can select one in 4 ways.

Hence the required number of ways = $6 + 4 = 10$.

Example 5: A person wants to leave station Q. There are 4 routes from station Q to P and 5 routes from Q to R. In how many ways can he travel from the station Q?

Solution: He can go from Q to P in 4 ways and Q to R in 5 ways. To go from Q to P and Q to R are independent to each other. Hence the person can leave station Q in $4 + 5 = 9$ ways.

3. Meaning of Permutation and Combination

Each of the different arrangements which can be made by taking some or all of a number of things is called a permutation. Note that in an arrangement, the order in which the things arranged is considerable i.e., arrangement AB and BA of two letters A and B are different because in AB, A is at the first place and B is at the second place from left whereas in BA, B is at the first place and A is at the second place.

The all different arrangements of three letters A, B and C are ABC, ACB, BCA, BAC, CAB and CBA.

Here each of the different arrangements ABC, ACB, BCA, BAC, CAB and CBA is a permutation and number of different arrangement i.e. 6 is the number of permutations.

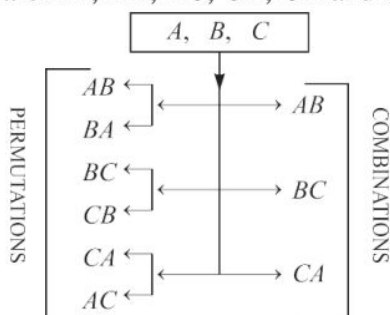
ABC, ACB, BCA, BAC, CAB and CBA are different arrangements of three letters A, B and C, because in each arrangement, order in which the letters arranged is considered. But if the order in which the things are arranged is not considered; then ABC, ACB, BCA, BAC, CAB and CBA are not different but the same. Similarly, AB and BA are not different but the same.

Each of the different selections or groups which can be made by some or all of a number of given things without reference to the order of things in any selection or group is called a combination. As in selection order in which things are selected is not considered; hence, selections of two letters AB and BA out of three letters A, B and C are the same. Similarly, selections of BC and CB are the same.

Also selections of CA and AC are the same.

Hence selection of two letters out of the three letters A, B and C can be made as AB, BC and CA only.

As in arrangements, order in which things are arranged is considered. Hence all arrangements of two letters out of the three letters A, B and C are AB, BA, BC, CB, CA and AC.



Number of permutations (or arrangements) of two letters out of three letters A, B and C = 6.

Number of combinations (or groups) of two letters out of three letters A, B and C = 3.

Permutations of three different letters A, B and C taken two at a time is also understood as selections of any two different letters AB, BC or CA out of A, B and C, then the selected two letters arranged in two ways as AB, BA; BC, CB or CA, AC

Example 6:

In how many ways can the letters of the word WATER be arranged so that we have a new pattern every time?

Solution:

This is permutation of n different things taken all at a time which is equal to $n!$

Hence, total number of different arrangements possible is $5! = 120$.

Another way to look at it is we have 5 places to be occupied by 5 different letters.

The 1st place can be filled by any of the 5 letters, hence 5 ways. The 2nd place can be filled by any one of the remaining 4 letters as one letter has already been fixed at the first place, hence 4 ways. Similarly, the 3rd place can be filled in 3 ways and the 4th in 2 ways. The 5th place can be filled in only one way as there is no choice but to fill it by the remaining 1 letter.

So going by the product rule, this can be done in $5 \times 4 \times 3 \times 2 \times 1 = 120$ ways.

Permutation of n things when some are identical or Permutation of n things not all different:

What happens when we have to find out the number of permutations when certain items are identical?

- If 2 exactly similar red chairs (R1 & R2) and 1 black chair (B) are to be arranged, then please note that one cannot distinguish between the 2 red chairs. This is to say that there is no difference between R1 B R2 and R2 B R1 because they will both look the same as I cannot differentiate between R1 and R2 as they are exactly same.
- So the total no. of arrangements possible will be $3! / 2! = 3$. They will be BRR RBR RRB.
- We have divided by $2!$ to take care of the 'two' items that are same.

If out of n things, p are exactly alike of one kind, q exactly alike of second kind and r exactly alike of third kind and the rest are all different, then the number of permutations of n things taken all at a time = $n! / (p!q!r!)$

Example 7:

In how many ways can the letters of the word COMMITTEE be arranged?

- using all the letters
- if all the vowels are together

Solution:

- Total letters = 9 and identical letters are 2M 2T and 2E.

So total no. of arrangements = $9! / 2!2!2!$

- Since all vowels must appear together we consider them as one unit.

There are 4 vowels- O I E E.

So now we have 5 letters. Out of these we have 2M and 2T. These 5 letters can be arranged in $5! / 2! 2!$ ways.

In the group of 4 vowels, the 4 vowels can arrange themselves in $4! / 2!$ ways.

So total no. of words formed = $5! / 2! 2! \times 4! / 2!$

Permutations where repetitions are allowed:

While dealing with letters and digits, you will often come across cases where repetition in permutation is allowed or not allowed. You have to be very careful as to what is asked for because the treatment for both the cases is absolutely different.

Example 8:

How many numbers of 5 digits can be formed with the digits 0,1,2,3,4?

- if the digits cannot repeat themselves
- if the digits can repeat themselves

Solution: